Summary in Graph

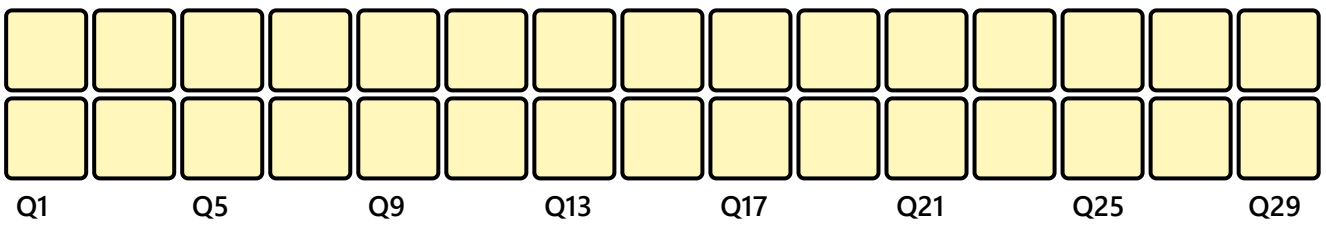
Exam Summary (GATE Overflow CS Test Series 2026 | Discrete Mathematics | Test 4)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	30 (10 + 20)
Total Marks:	50 (10 + 40)
Exam Duration:	90 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (3) out of 32 users

EXAM RESPONSE | EXAM STATS | FEEDBACK

Technical

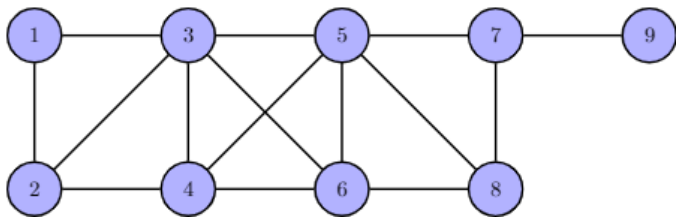
Q #1 | Numerical Type | Award: 1 | Penalty: 0 | Graph Theory

If G is a simple graph with 16 edges and $\overline{G}$ has 12 edges, how many vertices does the complement graph $\overline{G}$ have?

Your Answer: | Correct Answer: 8 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 Numerical Type | Award: 1 | Penalty: 0 | Graph Theory

The size of the minimum vertex cover of the below graph is _____



Your Answer: | Correct Answer: 5 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 Numerical Type | Award: 1 | Penalty: 0 | Graph Theory

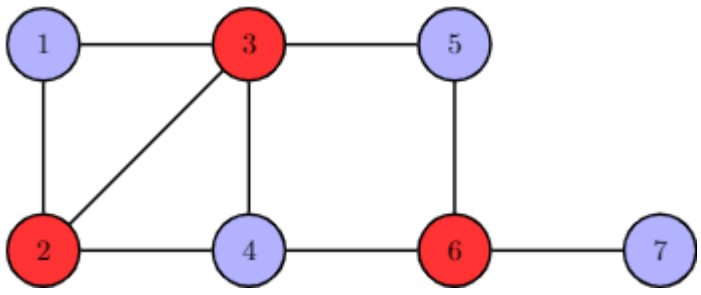
The number of possible connected simple graphs with 3 labelled vertices is _____

Your Answer: | Correct Answer: 4 | Not Attempted | Time taken: 00min 00sec | Discuss

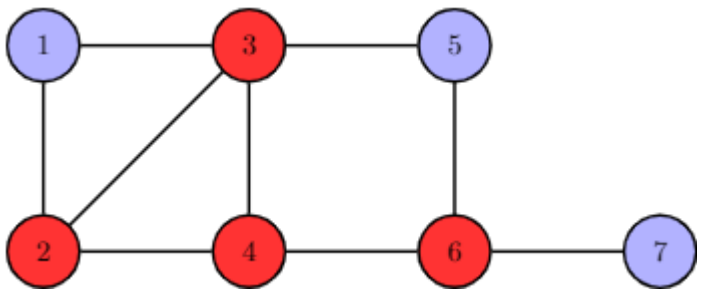
Q #4 Multiple Select Type | Award: 1 | Penalty: 0 | Graph Theory

In the graphs given below certain vertices are marked in RED. Which of the marked vertices form a vertex cover? (Mark all the appropriate choices)

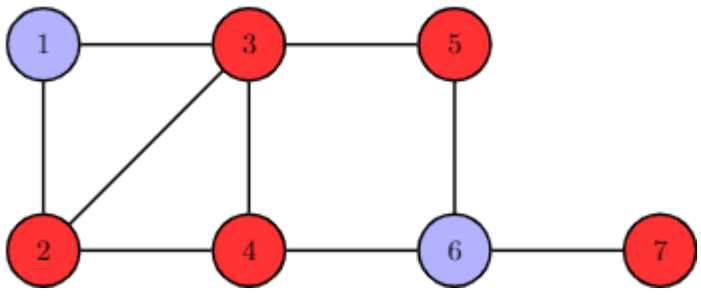
A.



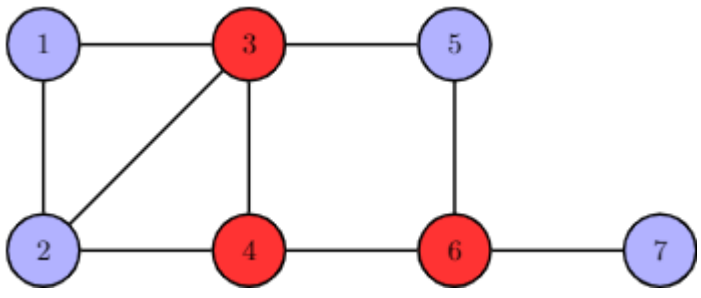
B.



C.



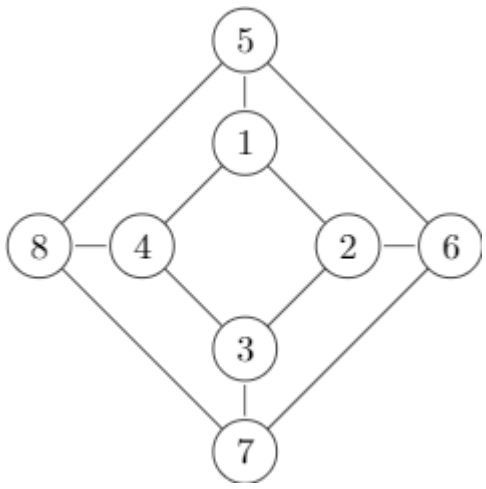
D.



Your Answer: | Correct Answer: A;B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 Numerical Type | Award: 1 | Penalty: 0 | Graph Theory

The minimum number of color required to color the following graph is _____



Your Answer: | Correct Answer: 2 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Graph Theory

Which of the following is/are correct?

S_1 : A connected graph of at least 2 vertices has an Euler circuit if and only if degree of every vertex is even.

S_2 : A connected graph has an Euler path if and only if there are at most two vertices with odd degree.

A. S_1 only
B. S_2 only
C. Both S_1 and S_2
D. Neither S_1 nor S_2

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Numerical Type | Award: 1 | Penalty: 0 | Graph Theory

Suppose a simple graph has 45 edges, 5 vertices of degree 6, and all others of degree 5. How many vertices does the graph have?

Your Answer: | Correct Answer: 17 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Numerical Type | Award: 1 | Penalty: 0 | Graph Theory

The maximum number of edges in a disconnected graph having 12 vertices is _____

Your Answer: | Correct Answer: 55 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Numerical Type | Award: 1 | Penalty: 0 | Graph Theory

What is the smallest number of colors that can be used to color the vertices of a cube so that no two adjacent vertices are colored identically?

Your Answer: | Correct Answer: 2 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Numerical Type | Award: 1 | Penalty: 0 | Graph Theory

The maximum number of edges a graph with 45 vertices, and 15 connected components is _____

Your Answer: | Correct Answer: 465 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Numerical Type | Award: 2 | Penalty: 0 | Graph Theory

In an academy 10 people are there and each one of them has an adversary and no one has more than one adversary. During a meeting all 10 of them shook hands with everyone else except their adversary. The number of handshakes that took place is _____

Your Answer: | Correct Answer: 40 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Graph Theory

Consider the following two statements with respect to a group of 9 people:

- S_1 : It is possible for everyone to be friends with exactly 2 others in the group
- S_2 : It is possible for everyone to be friends with exactly 3 others in the group

Which of the above two statements is/are TRUE?

A. S_1 only
B. S_2 only
C. Both S_1 and S_2
D. Neither S_1 nor S_2

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Multiple Select Type | Award: 2 | Penalty: 0 | Graph Theory

Which of the following statements concerning K_n- (a complete graph on n vertices) is/are TRUE? (Mark all the appropriate choices)

A. K_5 contains an Euler circuit
B. K_4 contains an Euler circuit
C. K_4 contains a Hamiltonian cycle
D. K_5 contains a Hamiltonian cycle

Your Answer: | Correct Answer: A;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Multiple Select Type | Award: 2 | Penalty: 0 | Graph Theory

Which of the following statements concerning $K_{m,n}-$ (a complete bipartite graph of partition sizes m and n) is/are TRUE? (Mark all the appropriate choices)

A. $K_{4,4}$ contains an Euler circuit
B. $K_{5,2}$ contains an Euler path
C. $K_{5,5}$ contains a Hamiltonian cycle
D. $K_{5,4}$ contains a Hamiltonian path

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Graph Theory

Which of the following is/are true?

- S_1 : The number of directed simple graphs with n-vertices is 2^{n^2-n}.
- S_2 : The number of undirected simple graphs with n-vertices is $2^{\binom{n}{2}}$.

A. Only S1
B. Only S2
C. Both S1 and S2
D. Neither S1 nor S2

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #16 Multiple Select Type | Award: 2 | Penalty: 0 | Graph Theory

Which of the following integer sequence if used as degree sequence would represent a simple graph? (Mark all the appropriate choices)

A. $(8, 8, 6, 5, 5, 5, 4, 4, 3, 3, 2, 2, 0)$
B. $(7, 7, 6, 5, 5, 4, 3, 3, 3, 3, 2, 1)$
C. $(1, 1, 1, 2, 2, 2, 3, 3, 4, 4, 4, 5, 5)$
D. $(7, 7, 6, 6, 5, 5, 4, 4, 3, 3, 2, 2, 1, 1)$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

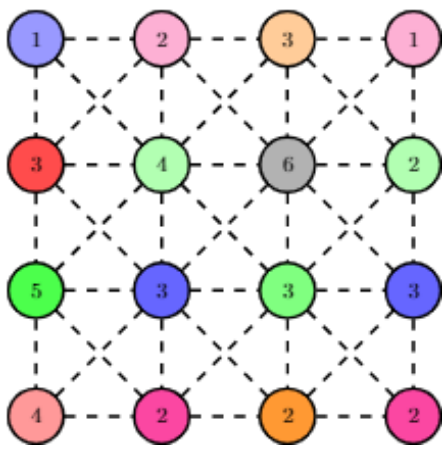
Q #17 Numerical Type | Award: 2 | Penalty: 0 | Graph Theory

What is the minimal number k such that there exists a proper edge coloring (chromatic index) of the complete graph on 6 vertices with k colors?

Your Answer: | Correct Answer: 5 | Not Attempted | Time taken: 00min 01sec | Discuss

Q #18 Numerical Type | Award: 2 | Penalty: 0 | Graph Theory

The project of building 42 roads connecting 16 cities is under way, as outlined below. So far, only some of the 42 roads are constructed, and the digit on each city indicates the number of constructed roads to other cities. How many complete roads are there among these cities?

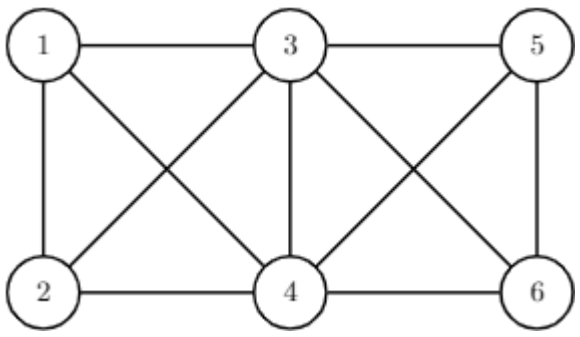


Your Answer: | Correct Answer: 23 | Not Attempted | Time taken: 00min 00sec | Discuss

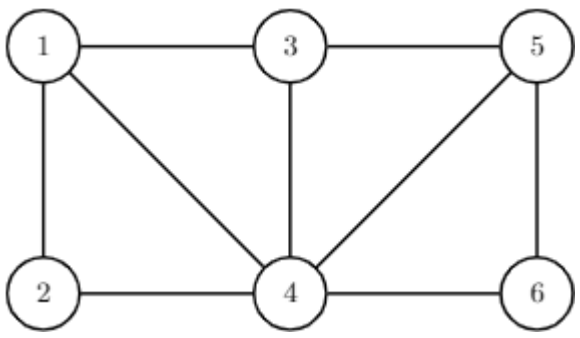
Q #19 | Multiple Select Type | Award: 2 | Penalty: 0 | Graph Theory

Which of the following graphs has an Euler path? (Mark all the appropriate choices)

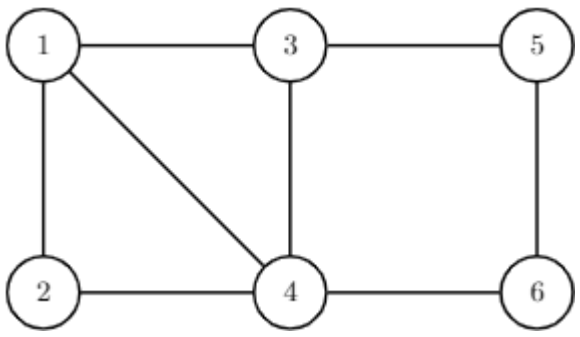
A.



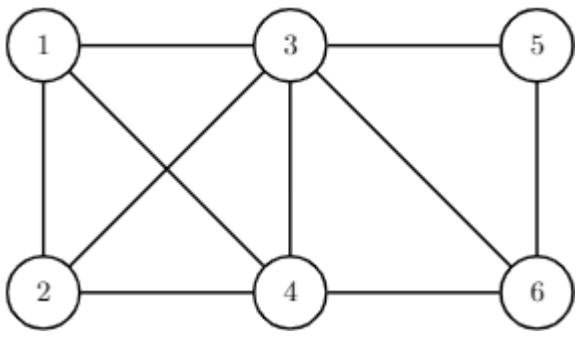
B.



C.



D.

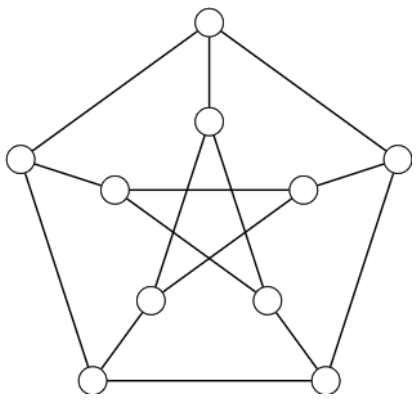


Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

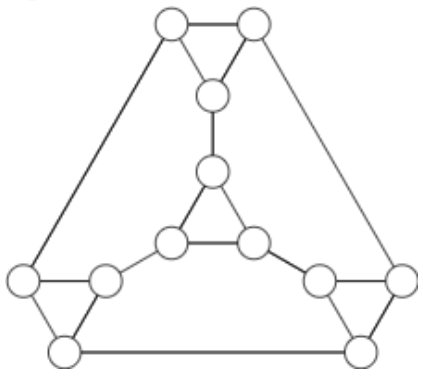
Q #20 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Graph Theory

Which of the following graphs has a Hamiltonian path?

S_1 :



S_2 :

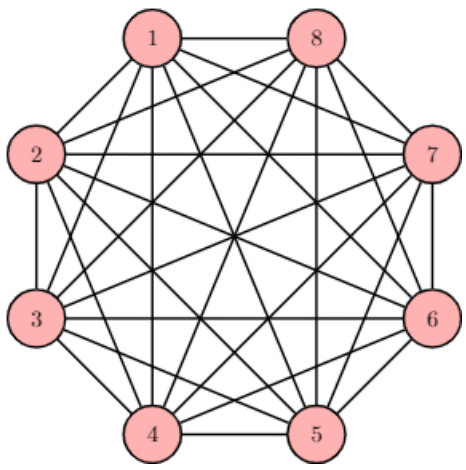


A. S_1 only
B. S_2 only
C. Both S_1 and S_2
D. Neither S_1 nor S_2

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #21 Numerical Type | Award: 2 | Penalty: 0 | Graph Theory

The number of Hamiltonian cycles on the following graph is _____



Your Answer: | Correct Answer: 2520 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #22 Numerical Type | Award: 2 | Penalty: 0 | Graph Theory

Number of perfect matching in a complete graph with 8 vertices is _____

Your Answer: | Correct Answer: 105 | Not Attempted | Time taken: 00min 00sec | Discuss

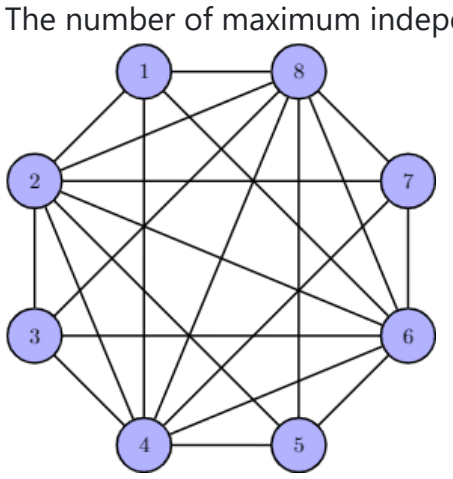
Q #23 Numerical Type | Award: 2 | Penalty: 0 | Graph Theory

The maximum number of edges in a bipartite graph with 19 vertices is _____

Your Answer: | Correct Answer: 90 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #24 Numerical Type | Award: 2 | Penalty: 0 | Graph Theory

The number of maximum independent vertex set of the below graph is _____

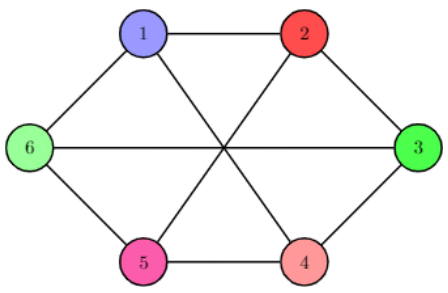


Your Answer: | Correct Answer: 1 | Not Attempted | Time taken: 00min 00sec | Discuss

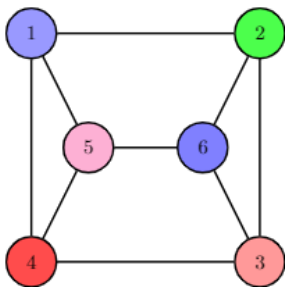
Q #25 | Multiple Select Type | Award: 2 | Penalty: 0 | Graph Theory

Which of the following graph is NOT bipartite? (Mark all the appropriate choices)

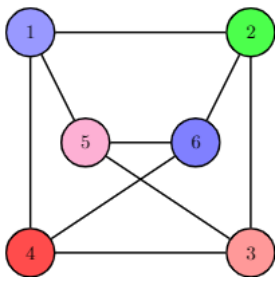
A.



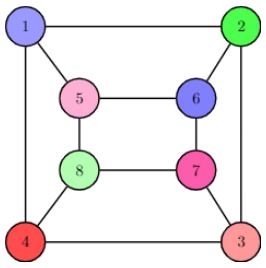
B.



C.



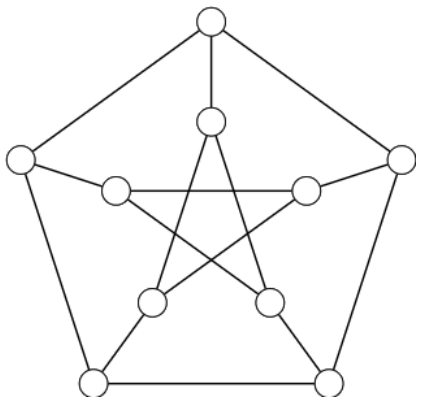
D.



Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #26 | Numerical Type | Award: 2 | Penalty: 0 | Graph Theory

The minimum number of color required to color the complement of the below graph is _____



Your Answer: | Correct Answer: 5 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #27 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Graph Theory

The chromatic numbers of the following graphs respectively are

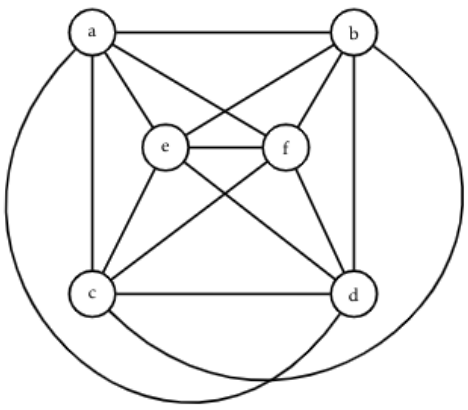
I. Complete graph K_n.
II. Complete bipartite graph $K_{m,n}$, where m and n are positive integers.
III. Cyclic graph C_n , where $n \geq 3$ and n is odd positive integers.
IV. Cyclic graph C_n , where $n \geq 4$ and n is even positive integers.

A. $n, \min(m, n), 2, 3$
B. $n - 1, 2, 3, 2$
C. $n, \max(m, n), 2, 3$
D. $n, 2, 3, 2$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #28 Numerical Type | Award: 2 | Penalty: 0 | Graph Theory

The edge chromatic number of the following graph is _____



Your Answer: | Correct Answer: 5 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #29 Multiple Select Type | Award: 2 | Penalty: 0 | Graph Theory

A connected graph G is chromatically $k-$critical if the chromatic number of G is k, but for every vertex of G, the chromatic number of the graph obtained by deleting this vertex from G is $k - 1$. Which of the following is/are correct? (Mark all the correct options) (C_n denotes the cyclic graph on n vertices and W_n denotes the wheel graph on n vertices)

A. C_n is chromatically $3-$critical whenever n is an odd positive integer, $n \geq 3$.
B. W_n is chromatically $3-$critical whenever n is an odd integer, $n \geq 3$.
C. W_n is chromatically $4-$critical whenever n is an even integer, $n \geq 4$.
D. If G is a chromatically $k-$critical graph, then the degree of every vertex of G is at least $k-1$.

Your Answer: | Correct Answer: A;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #30 | Numerical Type | Award: 2 | Penalty: 0 | Graph Theory

The number of perfect matching in $K_{7,7}$ is _____

Your Answer: | Correct Answer: 5040 | Not Attempted | Time taken: 00min 00sec | Discuss

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